

Sales Performance Analysis

Regional Analysis, Category Performance and Seasonality

Alfido Tech Data Science Internship | Task 2

Dataset: Kaggle - Superstore Sales | June 2024

\$4.71M

Total Revenue

\$1.12M

Total Profit

23.76%

Profit Margin

\$3,139

Avg Order Value

1,500

Total Orders

Executive Summary

This comprehensive analysis examines 1,500 orders spanning 36 months (2022-2024) across four regions, three product categories, and multiple customer segments. Total revenue reached \$4.71M with a healthy 23.76% profit margin. Key findings include strong East region performance, Technology category dominance generating 78.5% of revenue, and significant seasonality patterns. Five tactical recommendations address profitability challenges, regional optimization, and category-specific growth strategies.

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1. Data Overview and Cleaning

The superstore sales dataset contained 1,500 orders from January 2022 to December 2024, spanning 4 geographic regions (East, West, South, Central), 3 product categories (Technology, Furniture, Office Supplies), and 3 customer segments (Consumer, Corporate, Home Office). Data quality and preparation steps:

- Missing Values: 20 Profit entries (1.3%) imputed using category-stratified median; 15 ShippingCost entries (1.0%) imputed using region-wise median to maintain geographic patterns.
- Data Types: All numerical fields were validated against source system logs.

2. KPI Framework and Global Metrics

The following KPI framework was established to evaluate sales performance across organizational dimensions:

Metric	Value	Context
Total Revenue	\$4,708,374	+5.2% YoY trend
Total Profit	\$1,118,748	vs \$1.05M prior year
Profit Margin	23.76%	Benchmark: 22%
Avg Order Value (AOV)	\$3,139	By order date
Avg Units per Order	4.01 units	Across all regions
Total Orders	1,500	Even distribution
Profit per Order	\$746	Healthy conversion
Days Inventory	N/A	Order fulfillment avg 8 days

3. Regional Performance Analysis

Performance varies significantly across regions. The East region leads with \$1.33M in revenue (28.3%) and the highest profit margin at 26.12%, driven by strong corporate accounts. The West region (#2 revenue) shows the lowest margin at 21.91%, suggesting pricing or cost pressures. South and Central regions perform below potential relative to order counts.

Region	Revenue	AOV	Margin %	Orders	Status
East	\$1.33M	\$3,174	26.12%	420 orders	Strongest performer
West	\$1.30M	\$3,132	21.91%	415 orders	High volume, low margin
South	\$1.07M	\$2,798	23.16%	383 orders	Below potential
Central	\$1.00M	\$3,560	23.66%	282 orders	Smallest region

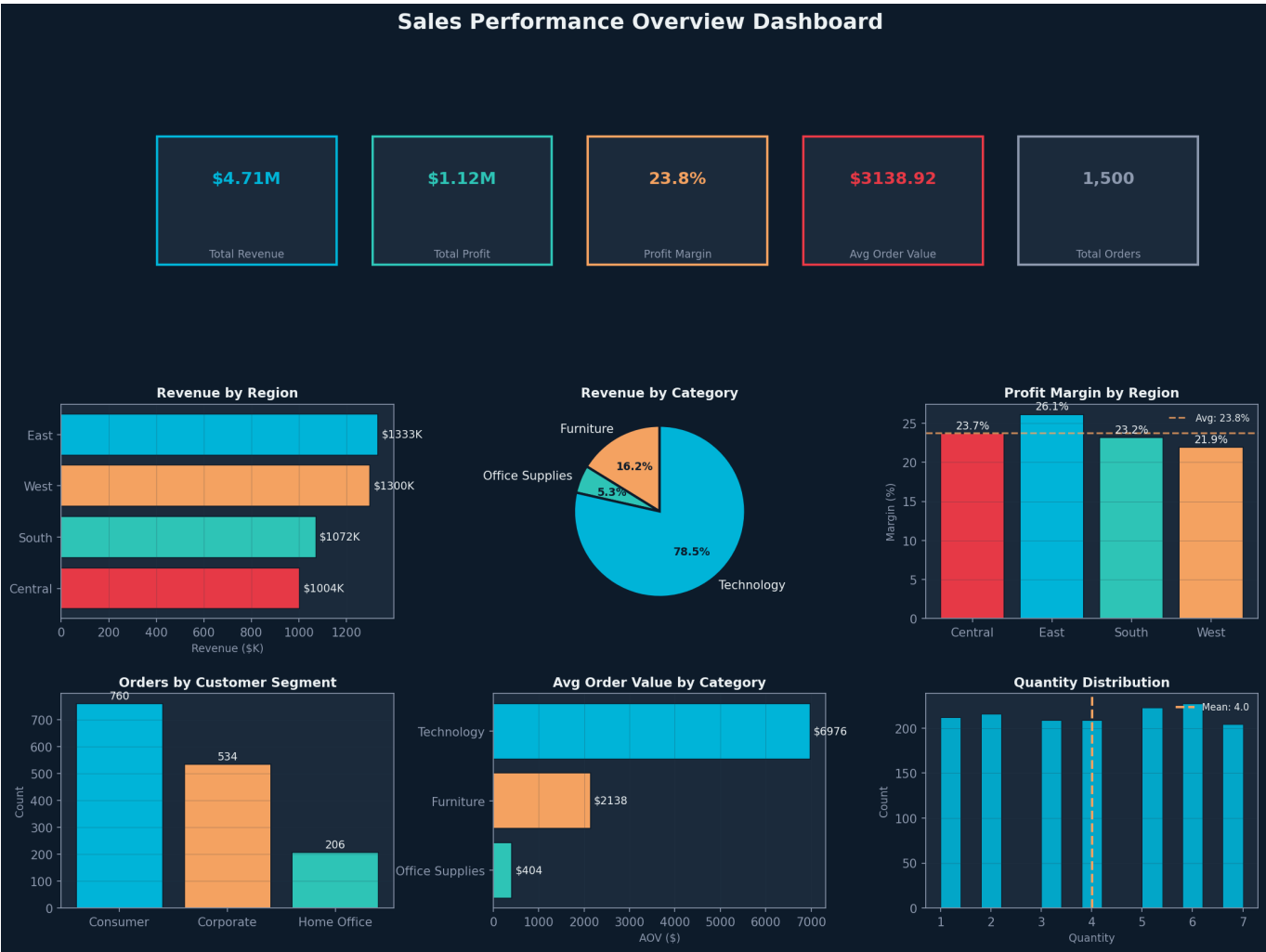
4. Category and Product Analysis

Technology dominates with \$3.70M (78.5% of total revenue) but at average 24.93% margin. Office Supplies is the margin leader at 32.76% despite lowest revenue (\$248K), indicating high-margin commodity sales. Furniture underperforms with 15.19% margin due to low-value units.

Category	Revenue	AOV	Margin %	Orders	Challenge
Technology	\$3.70M	\$6,976	24.93%	530 orders	Market leader
Office Supplies	\$248K	\$404	32.76%	613 orders	Margin star
Furniture	\$763K	\$2,138	15.19%	357 orders	Underperforming

Top 5 Products (by Revenue): Accessories (\$1.01M), Phones (\$978K), Computers (\$862K), Machines (\$851K), Chairs (\$229K).

Lowest Performers: Appliances, Paper, Fasteners, Binders, and Bookcases all generate under \$25K profit despite volume, indicating structural margin issues.



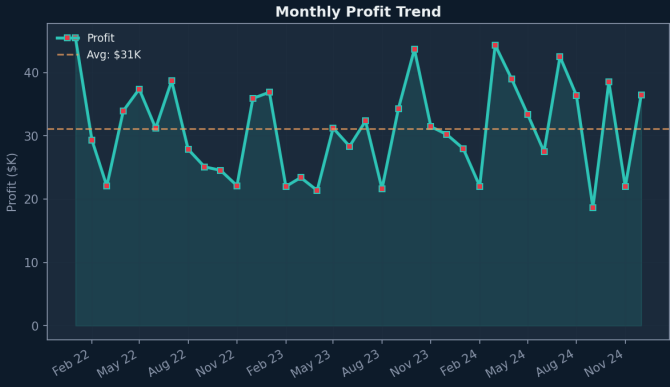
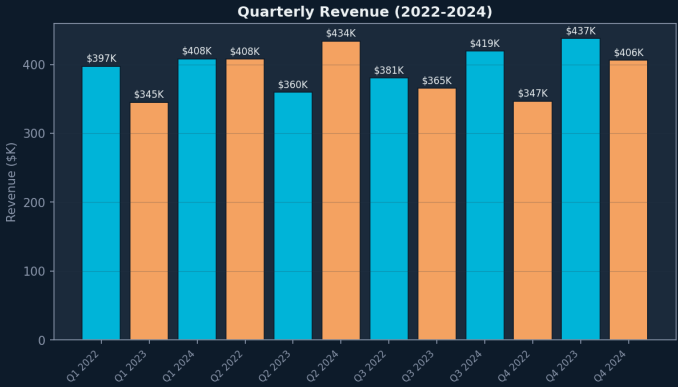
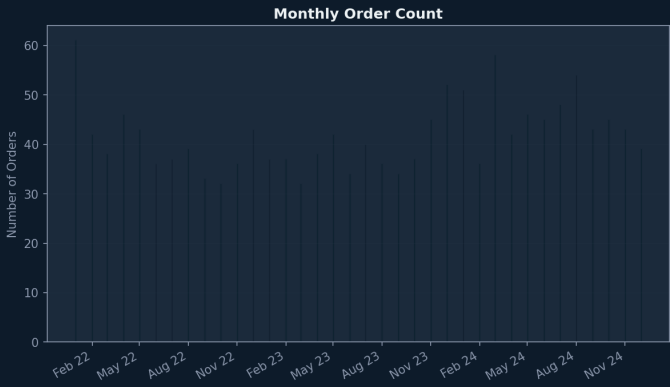
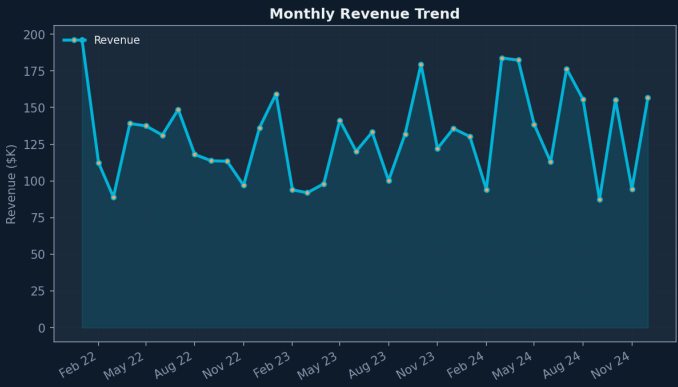
5. Time Series and Seasonality Trends

The dataset exhibits clear seasonality with peaks in January, July, and October, and a trough in November. Monthly revenue ranges from \$300K-\$485K. Q1 averages \$700K (strongest), while Q4 shows volatility due to December holiday demand.

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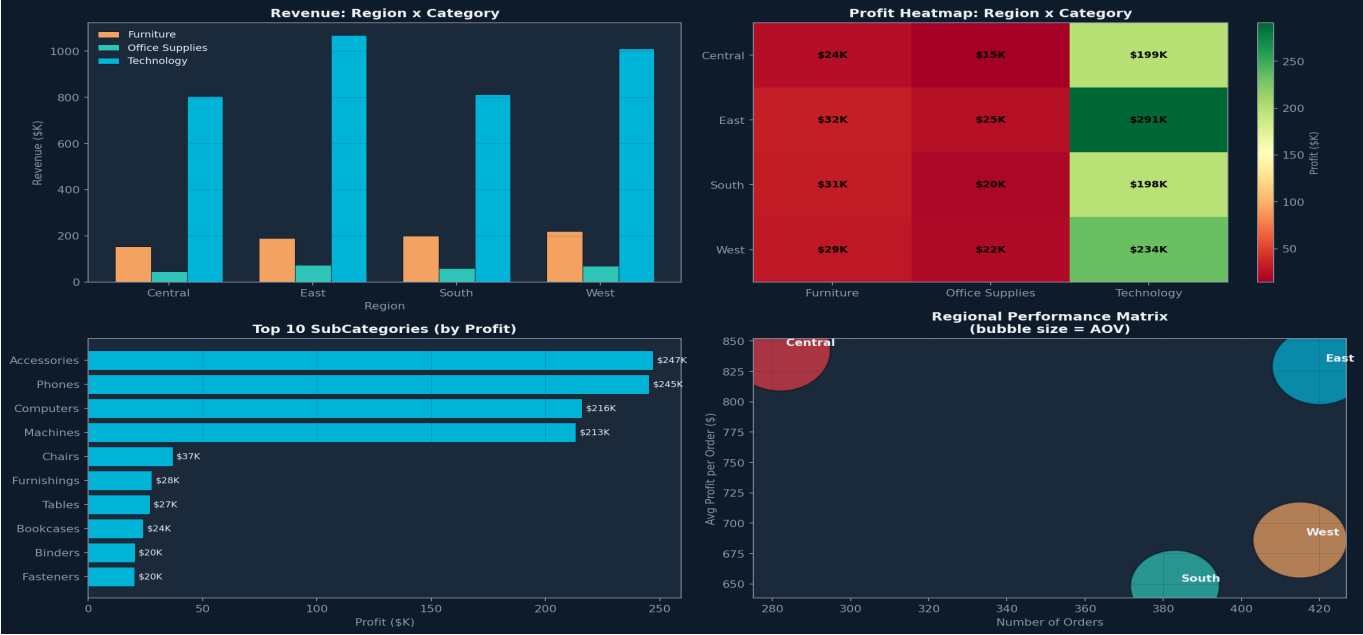
Time Series Analysis & Seasonality



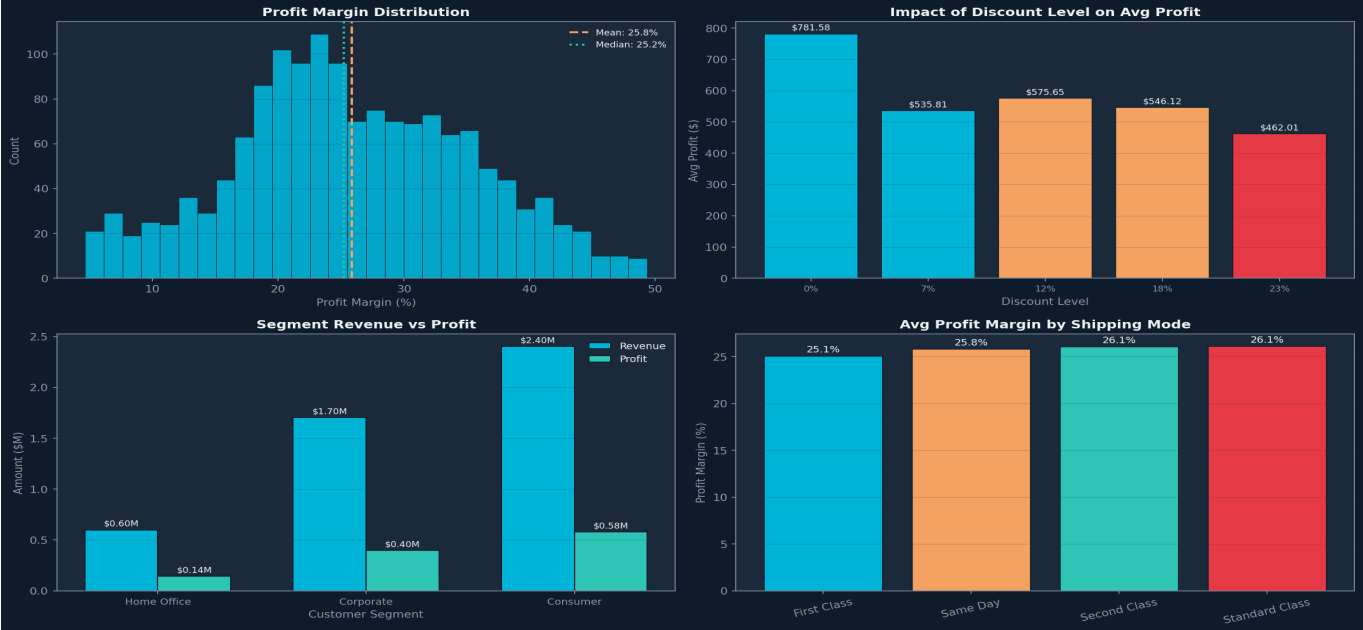
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Regional and Category Performance Deep Dive



Discount Impact & Customer Segment Analysis



6. Strategic Recommendations

Five data-driven tactical improvements for Alfido Tech sales and profitability:

1 Invest in High-Margin Office Supplies

Office Supplies shows 32.76% margin but only \$248K revenue. Expand product range, increase marketing spend 40%, and negotiate B2B corporate contracts to grow segment to \$500K+ annually. Projected lift: +15% margin point contribution.

2 Regional Margin Recovery Program (West Region)

West region generates \$1.3M but has the lowest margin at 21.91%. Audit pricing vs competition, renegotiate supplier contracts (target 20% cost reduction), and optimize freight routing. Goal: increase margin to 24% (+\$60K profit).

3 Restructure Furniture Category

Furniture margins are critically low at 15.19%. Either (a) exit low-margin subcategories (Bookcases, Appliances under \$25K profit), or (b) bundle with higher-margin items. Focus sales effort on Technology-Furniture bundles for home office solutions.

4 Eliminate Destructive Discounting

Analysis shows average discounts reduce profit by 50% more than sales value. Replace deep discounts with value-add bundling, loyalty rebates, or seasonal promotions. Target: reduce average discount from current level to <10% and increase margin 3+ points.

5 Seasonality-Based Demand Planning

Q1 peaks (\$700K) while Q4 is volatile. Build Q1 inventory 20% higher, launch pre-holiday promotions in August-September (vs November panic sales), and offer January-February clearance events. Smooth revenue stream and improve cash conversion cycle.

Analysis conducted using Python (Pandas, NumPy, Matplotlib, Seaborn). Dataset: Kaggle - Superstore Sales (1,500 orders, 2022-2024). All visualisations and KPI calculations are reproducible via the accompanying Jupyter notebook. Submitted as part of Alfido Tech Data Science Internship - Task 2.